practice_test1

This is a preview of the published version of the quiz

Started: Apr 7 at 12:25pm

Quiz Instructions

Question 1 11.2 pts

The current produced by voltage in a circuit is impeded by

- none of the above
- electric resistance.
- electric barriers.
- closed switches.

Question 2 11.2 pts

The resistance of a filament that carries 2 A when a 10-V potential difference across it is

- 5 ohms.
- 10 ohms.
- 20 ohms.
- more than 20 ohms.
- 2 ohms.

Question 3 11.2 pts

Four unequal resistors are connected in series with each other. Which one of the following statements is correct about this combination?

- The equivalent resistance is equal to that of any one of the resistors.
- The equivalent resistance is equal to average of the four resistances.

○ The equivalent resistance is less than that of the smallest resistor.

○ The equivalent resistance is less than that of the largest resistor.

○ The equivalent resistance is more than the largest resistance.

Question 4 11.2 pts

The power dissipated in a 4-ohm resistor carrying 3 A is

○ 18 W.

○ 7 W.

○ need more information

○ 48 W.

○ 36 W.

Question 5 11.2 pts

Four unequal resistors are connected in a parallel with each other. Which one of the following statements is correct about this combination?

○ The equivalent resistance is equal to the average of the four resistances.

○ The equivalent resistance is less than that of the smallest resistor.

○ None of the other choices is correct.

○ The equivalent resistance is more than the largest resistance.

○ The equivalent resistance is midway between the largest and smallest resistance.

Question 6 11.2 pts

A 6.0-Ω and a 12-Ω resistor are connected in parallel across an ideal 36-V battery. What power is dissipated by the 6.0-Ω resistor?

○ 48 W

○ 490 W

○
220 W

○
24 W

Question 7 11.2 pts

Four resistors having resistances of 20 Ω, 40 Ω, 60 Ω, and 80 Ω are connected in series across an ideal dc voltage source. If the current through this circuit is 0.50 A, what is the voltage of the voltage source?

○
80 V

○
20 V

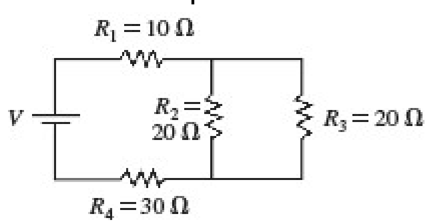
○
40 V

○
60 V

○
100 V

Question 8 11.2 pts

What is the equivalent resistance in the circuit shown in the figure?



○
55 Ω

○
50 Ω

○
80 Ω

○
35 Ω

Question 9 11.2 pts

A battery you buy at the store has an internal emf of 3.0 V. If it has an internal resistance of 14 ohms what current will this battery put out if it is short-circuited?

○
0.21 A

- 4.7 A
- 42 A
- 130 A

Question 10 1 pts

from MCAT

If a defibrillator passes 15 A of current through a patient's body for 0.1 seconds, how much charge goes through the patient's skin?

- 15C
- 0.5C
- 1.5C
- 150C

Question 11 1 pts

we did it for a popquiz. Build it with the simulation. MCAT

-

A student places an ammeter with negligible resistance in parallel with a resistor to determine the amount of current that passes through the resistor. Does the student obtain an accurate reading?

- **A.** Yes, because he used an ammeter with negligible resistance.
- **B.** Yes, because the current going through parallel paths is equal.
- **C.** No, because the ammeter should have infinite resistance.
- **D.** No, because an ammeter in parallel changes the current through the resistor.

- ◯ D
- ◯ C
- ◯ B
- ◯ A

Question 12 1 pts

MCAT. REview the relationship for connecting wires.

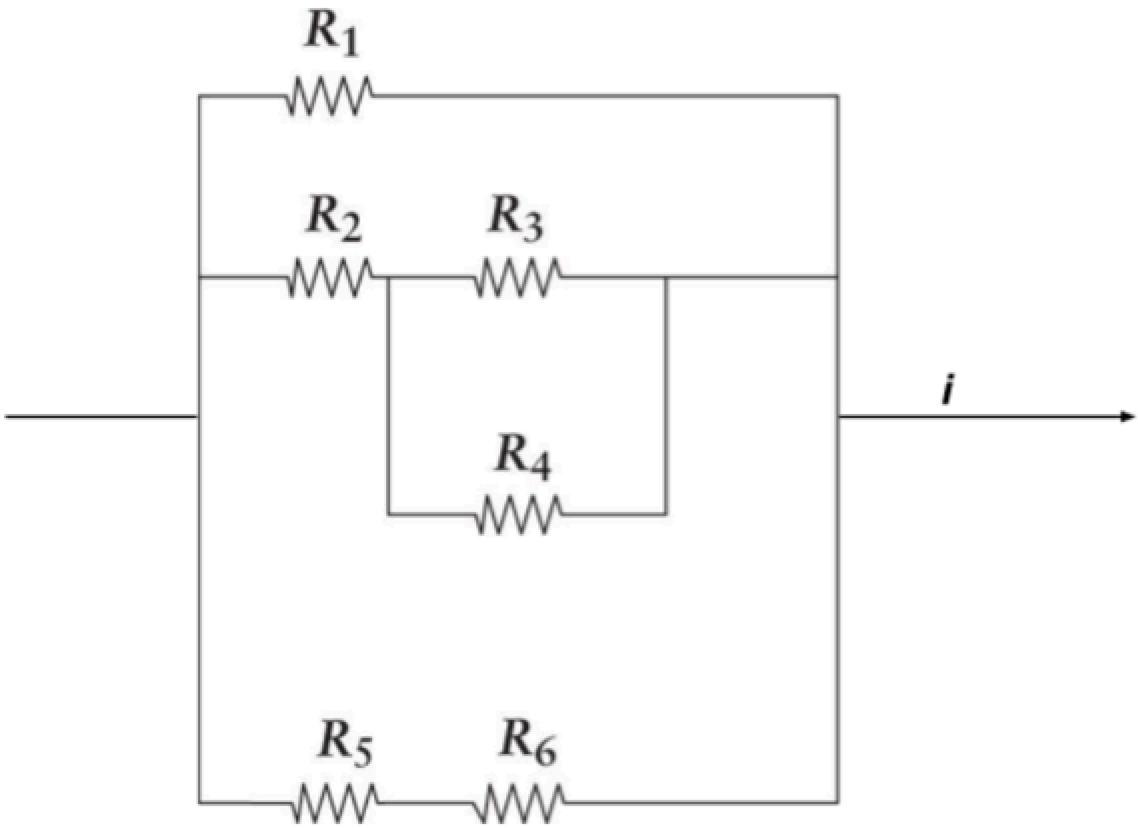
Question 3

The resistance of two conductors of equal cross-sectional area and equal lengths are compared, and are found to be in the ratio 1:2. The resistivities of the materials from which they are constructed must therefore be in what ratio?

- ◯ 1:1
- ◯ 1:2
- ◯ 4:!
- ◯ 2:1

Question 13 1 pts

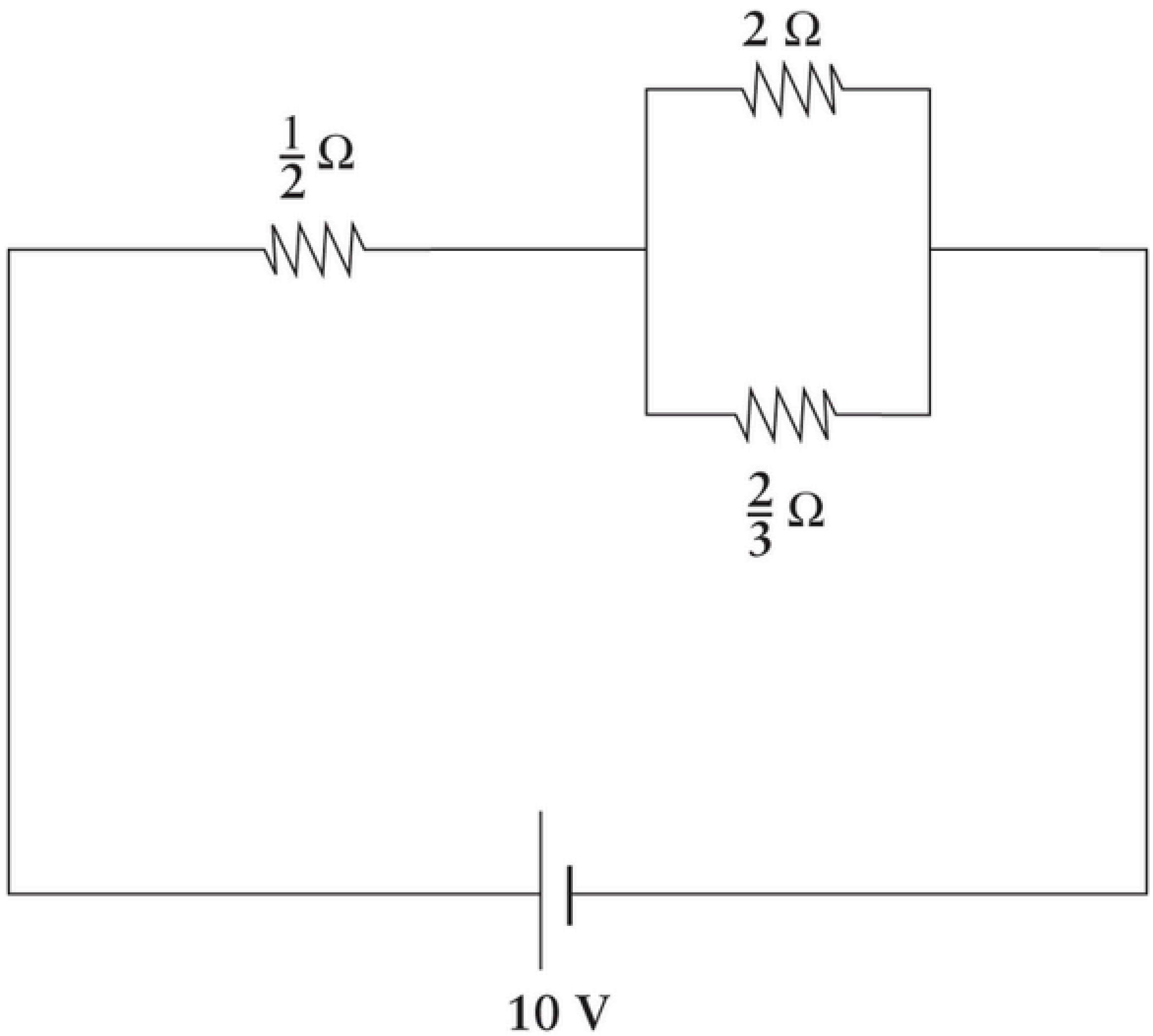
Given that R_1 = 20 Ω, R_2 = 4 Ω, R_3 = R_4 = 32 Ω, R_5 = 15 Ω, and R_6 = 5 Ω, what is the tot resistance in the circuit element shown below?



- 6.67 ohms
- 16.7 ohms
- 60 ohms
- 0.15 ohms

Question 14 1 pts

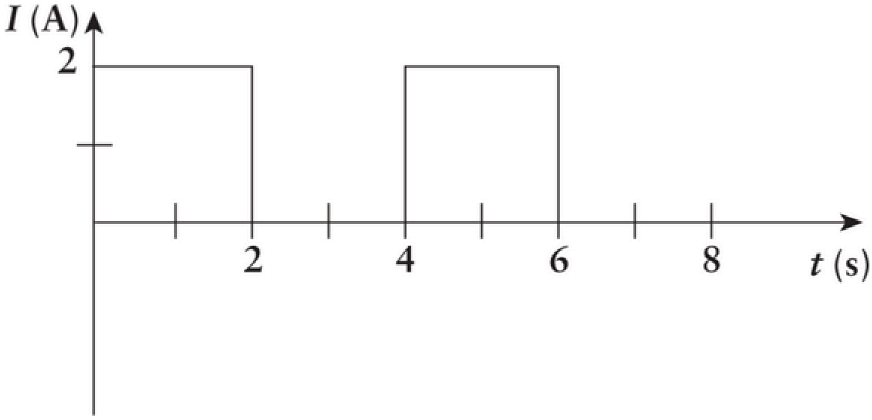
In the circuit below, what is the voltage drop across the $\frac{2}{3}\Omega$ resistor?



- ○ 2/3V
- ○ 1/2V
- ○ 7.5V
- ○ 5V

Question 15 1 pts

A 10 Ω resistor carries a current that varies as a function of time as shown. How much energy has been dissipated by the resistor after 5 s?



- A. 40 J
- B. 50 J
- C. 120 J
- D. 160 J

Tricky question . hide hint. (MCAT)

So total energy between 0<t<5s (so after 5 s using the circuit)

hint:. use P = RI^2

- 80J
- 40J
- 120J
- 160J

Question 16 1 pts

Which of the following best characterizes ideal voltmeters and ammeters?

- **A.** Ideal voltmeters and ammeters have infinite resistance.
- **B.** Ideal voltmeters and ammeters have no resistance.
- **C.** Ideal voltmeters have infinite resistance, and ideal ammeters have no resistance.
- **D.** Ideal voltmeters have no resistance, and ideal ammeters have infinite resistance.

MCAT

- D
- C
- B
- A

Not saved Submit Quiz